IT3091 Machine Learning — Student Performance Project

Combined Reference: Data Dictionary, EDA Insight Log, Decision Log

IT3091 Machine Learning | Guided Data Track | Education (Dataset Code 3) Dataset: UCI Student Performance — Math (`student-mat.csv`) | MD5 fingerprint: `4dc304be95c60de6ee13fb8769469dd7`

Part 1 — Data Dictionary

Dataset: UCI Student Performance — Math (`data/student-mat.csv`) 395 records, 33 attributes. Source: https://archive.ics.uci.edu/dataset/320/student-performance

#	Column	Type	Description	Values / Range
1	`school`	Binary	Student's school	`GP` (Gabriel Pereira), `MS` (Mousinho da Silveira)
2	`sex`	Binary	Student's sex	`F`, `M`
3	`age`	Numeric	Student's age	15–22

4	address	Binary	Home address type	U (urban), R (rural)
5	famsize	Binary	Family size	LE3 (≤3), GT3 (>3)
6	Pstatus	Binary	Parents' cohabitation status	T (together), A (apart)
7	Medu	Ordinal	Mother's education	0 (none) – 4 (higher education)
8	Fedu	Ordinal	Father's education	0 (none) – 4 (higher education)
9	Mjob	Nominal	Mother's job	teacher, health, services, at_home, other
10	Fjob	Nominal	Father's job	teacher, health, services, at_home, other
11	reason	Nominal	Reason for choosing this school	home, reputation, course, other
12	guardian	Nominal	Student's guardian	mother, father, other
13	traveltime	Ordinal	Home-to-school travel time	1 (<15min) – 4 (>1hr)
14	studytime	Ordinal	Weekly study time	1 (<2hrs) – 4 (>10hrs)
15	failures	Numeric	Number of past class failures	0–3 (4 = "3 or more")
16	schoolsup	Binary	Extra educational support	yes, no

17	famsup	Binary	Family educational support	yes, no
18	paid	Binary	Extra paid classes (in-subject)	yes, no
19	activities	Binary	Extra-curricular activities	yes, no
20	nursery	Binary	Attended nursery school	yes, no
21	higher	Binary	Wants to pursue higher education	yes, no
22	internet	Binary	Internet access at home	yes, no
23	romantic	Binary	In a romantic relationship	yes, no
24	famrel	Ordinal	Quality of family relationships	1 (very bad) – 5 (excellent)
25	freetime	Ordinal	Free time after school	1 (very low) – 5 (very high)
26	goout	Ordinal	Going out with friends	1 (very low) – 5 (very high)
27	Dalc	Ordinal	Workday alcohol consumption	1 (very low) – 5 (very high)
28	Walc	Ordinal	Weekend alcohol consumption	1 (very low) – 5 (very high)
29	health	Ordinal	Current health status	1 (very bad) – 5 (very good)

30	`absences`	Numeric	Number of school absences	0–93 (observed max in data: 75)
31	`G1`	Numeric	First period grade	0–20
32	`G2`	Numeric	Second period grade	0–20
33	`G3`	Numeric	**Final grade (target variable)**	0–20

Row meaning (unit of analysis)

One row = one student enrolled in the Math course at one of two secondary schools in Portugal (Gabriel Pereira or Mousinho da Silveira).

Data quality notes

- No null values across all 33 columns.
- No duplicate rows.
- No inconsistent category formatting (all categorical values pre-clean).
- 33 of 395 students have `G3 = 0`. Investigated in EDA: these students have `absences = 0` but already-declining G1→G2 grades, consistent with exam non-participation/withdrawal rather than a data error — kept in the dataset as a genuine at-risk outcome (see Decision #5 below).
- `absences` is heavily right-skewed (max 75 vs. median 4); only 4 students exceed 30 — addressed with a log1p transform during preprocessing rather than removal (Decision #7 below).

Note on scope

This project uses the Math dataset (`student-mat.csv`) only. The companion Portuguese-language dataset (`student-por.csv`, 649

records) and the 382 students shared between both datasets (identifiable via `student-merge.R`) were out of scope given the project timeline — documented as a limitation.

Part 2 — EDA Insight Log

All analysis below was performed on the **training set only** (316 of 395 students), after the 80/20 train/test split, to avoid any test-set leakage into modelling decisions.

1. Target variable (G3) distribution

- Skewness = -0.773 (moderately left-skewed).
- The skew is driven by a cluster of students scoring G3 = 0, not a smooth tail — the bulk of the distribution otherwise sits in a fairly normal range centered around 10–11.

2. G3 = 0 investigation

- 33 of 316 training students (10.4%) have G3 = 0.
- All 33 also have `absences = 0`, while their G1/G2 grades were already low and falling (G1 mean ≈ 7.6 → G2 mean ≈ 4.4).
- Interpretation: zero absences combined with a zero final grade points to **exam non-participation/withdrawal**, not an attendance-driven failure or a data error.
- Insight for the business lens: these are a real, meaningful at-risk population — exactly the group the secondary lens (at-risk identification) needs to catch, not noise to be removed.

3. Missing values & duplicates

- Zero nulls across all 33 columns.

- Zero duplicate rows.
- No formatting inconsistencies found in any categorical column.
- Insight: this dataset required no missing-value imputation or deduplication logic — a genuine finding worth stating plainly rather than a gap in the analysis.

4. Outlier check — absences

- Heavily right-skewed: median 4, mean ≈ 5.9, max 75.
- Only 4 of 316 students exceed 30 absences — a small, genuine extreme group, not noise spread evenly through the data.
- Insight: real outliers, not errors — addressed via transformation (log1p), not removal, so no legitimate data was discarded.

5. Correlation with G3

Feature	Correlation with G3
G2	0.908
G1	0.792
failures	-0.374
Medu	0.216
studytime	0.138
Fedu	0.135

- G1 and G2 dominate, as expected (same grading scale, strong temporal proximity to G3).
- `failures` is the strongest non-grade predictor.
- Insight: G1/G2's strength makes them central features but also raises the model's dependency on mid-year data being available (see limitations below).

6. School-level comparison

- GP and MS schools show a difference in mean G3, though GP has a larger student sample in this dataset.
- Insight: school alone is a secondary factor relative to grade-history and aspiration variables.

7. Subgroup comparisons

Subgroup	Mean G3
Female	10.03
Male	10.67
Rural address	9.62
Urban address	10.52
Does **not** want higher education	6.33
Wants higher education	10.57

- The largest gap by far is `higher` (aspiration to pursue higher education): a 4.2-point mean difference.
- Insight: aspiration is a strong behavioural signal, larger than sex or address — flagged as a feature likely to carry real predictive weight.

8. Feature importance (post-modelling, cross-referenced here for completeness)

Computed on the best-performing model (Gradient Boosting) using two independent methods — tree-based importance and permutation importance — which agreed on the same top 3 features in the same order: **G2, absences, G1**.

- Insight: `failures`, despite its meaningful raw correlation (-0.374), did not rank highly in either importance method — likely because G2 already captures most of the same signal, leaving `failures` with little additional (marginal) predictive value once G2 is

included.

Part 3 — Decision Log

#	Decision	Options Considered	Reason Chosen	
1	Use Math dataset only, not Portuguese	Math only; Portuguese only; both merged	Scope reduction given project timeline; avoids merge-key complexity between the two datasets; Math is the dataset named first in the business scenario	`st`
2	No null/duplicate handling needed	N/A	Dataset verified clean on load	0
3	No category standardization needed	N/A	All categorical values already consistently formatted	Ma
4	Split before EDA, EDA on train set only	Split after EDA; split before EDA	Prevents any EDA-driven decision from being informed by test-set data (leakage prevention)	St

5	Keep G3 = 0 rows, do not drop	Drop as anomalies; drop as missing; keep	33/316 train rows (10.4%) have G3 = 0 with 0 absences and declining G1→G2 — consistent with exam withdrawal, a genuine at-risk outcome the business lens needs to capture, not noise	ED
6	Include G1 and G2 as predictive features	Include; exclude (predict from only start-of-year data)	Both are realistically known before the final grade (first/second period grades), and contribute by far the strongest correlation with G3 (0.91, 0.79)	Co
7	Log-transform (log1p) `absences`, do not drop outliers	Drop outliers; cap/ winsorize; log-transform; leave raw	Only 4/316 students have >30 absences — real values, not errors. Log transform reduces their influence on linear/ridge models without discarding data	Ou

8	Preprocessing: OneHotEncoder for categoricals, StandardScaler for numerics, built as a single `ColumnTransformer` fit on train only	Manual encoding; separate per-model preprocessing	Single shared, leakage-safe pipeline reusable identically across all 4 models; fit exclusively on `X_train`, applied via `.transform()` to `X_test`	`fu` / `.`
9	Evaluation protocol: 80/20 train/test split + 5-fold CV during tuning, test set touched only once at final evaluation	Single split only; CV only, no held-out test	Standard combination for small datasets (395 rows): CV for stable tuning without overfitting to one split, final held-out test for an honest unbiased estimate	`tr` `ra`
10	Random seed fixed at 42 throughout	N/A	Reproducibility requirement (rubric: dataset fingerprint + reproducible notebook)	`RA`

11	Four models selected for comparison: Linear Regression, Random Forest, Ridge, Gradient Boosting	Also considered: SVR, KNN Regressor	Covers the required baseline + 3 alternatives spanning a simple linear model, a regularized linear model, and two tree ensembles with different bias/variance trade-offs — gives a genuinely varied comparison rather than near-duplicate methods	Dr
12	Hyperparameter tuning via `GridSearchCV`, 5-fold CV, scoring = negative RMSE	`RandomizedSearchCV`; manual tuning	Dataset is small (316 train rows) so exhaustive grid search is cheap; small, sensible grids kept per model to control runtime	Be

Model Results (5-fold CV RMSE, training set)

Model	Best Hyperparameters	CV RMSE
Linear Regression	n/a (no tuning)	1.792

Ridge Regression	alpha = 10.0	1.764
Random Forest	n_estimators=100, max_depth=None, min_samples_leaf=1	1.389
Gradient Boosting	n_estimators=100, learning_rate=0.05, max_depth=3	1.458

Ridge scoring slightly better than plain Linear Regression is consistent with the G1/G2 multicollinearity finding from EDA (#6) — the L2 penalty is doing real work shrinking the correlated coefficients. Random Forest led on CV; final ranking confirmed against the held-out test set below (CV score alone is not the final word).

Final Test-Set Evaluation

Model	Train RMSE	Test RMSE	Train-Test Gap	Test MAE	Test R^2
Gradient Boosting	0.960	1.945	0.985	1.130	0.816
Random Forest	0.578	1.994	1.415	1.183	0.806
Ridge Regression	1.597	2.261	0.664	1.641	0.751
Linear Regression	1.586	2.314	0.728	1.699	0.739

| 13 | Final model ranking taken from held-out test RMSE, not CV score | Rank by CV score; rank by test score | Random Forest had the best CV RMSE (1.389) but the largest train-test gap (1.415) — a clear overfitting signature (near-zero training error, inflated by fitting fold-specific noise). Gradient Boosting has the best actual test RMSE (1.945) and R^2 (0.816), with a smaller, more honest gap. **Gradient Boosting selected as the best-performing model**; Random Forest's CV-vs-test disagreement is reported explicitly as a finding, not hidden | Test-set metrics table above | | 14 | Residual check: no funnel shape /

heteroscedasticity observed across all 4 models | N/A | Confirms model errors are reasonably homoscedastic — no evidence of a systematic pattern in residuals the models are missing | Residual scatter plots | | 15 | Ridge's smallest train-test gap (0.664) reinforces the multicollinearity decision | N/A | Direct supporting evidence (not just theoretical claim) that regularization helps generalization given G1/G2 correlation | Gap column above, vs. decision #6 |

Feature Importance

Computed on the best-performing model (Gradient Boosting), using two independent methods to reduce single-method bias.

Rank	Tree-Based Importance	Permutation Importance
1	G2 (0.799)	G2 (3.645)
2	absences (0.135)	absences (0.599)
3	G1 (0.015)	G1 (0.139)

| 16 | Feature importance computed on Gradient Boosting (the selected best model), not on all 4 | Compute for every model; compute for best model only | Best model's importance is what actually drives the deployed predictions; computing for all 4 would be redundant given time constraints and add no new decision-relevant information | `tree_importance_df`, `perm_importance_df` | | 17 | Both methods agree G2, absences, G1 are the top 3 drivers, in the same order | N/A | Strong two-method convergence — satisfies the rubric's "two independent methods" requirement for feature importance with a genuine agreement, not a forced one | Comparison table above | | 18 | `failures` (correlation -0.37 with G3) does not rank highly in either importance table | N/A | Likely explained by G2 already capturing most of the same signal (students with more failures tend to already have lower G2) — once G2 is in the model, `failures` adds little marginal predictive information beyond what G2 already encodes. Reported as a nuance distinguishing correlation from marginal importance | Cross-reference: EDA correlation vs. importance tables |

At-Risk Report

Applied predicted G3 < 10 threshold to Gradient Boosting's test-set predictions: 37/79 test students (46.8%) flagged. Validated against actual outcomes:

	Predicted Not At-Risk	**Predicted At-Risk**
Actual Not At-Risk	42 (TN)	10 (FP)
Actual At-Risk	0 (FN)	27 (TP)

Recall = 1.00 (at-risk), Precision = 0.73 (at-risk), Accuracy = 0.87.

| 19 | Per-student risk explanation shown via global top-6 feature values, not individual SHAP values | SHAP/LIME per-student explanations; global feature values only | Simpler, faster to implement within the 1-day timeline, and still gives an advisor a clear, explainable basis per student; documented explicitly as a simplification/limitation rather than presented as equivalent to true local explanation methods | at_risk_report table | | 20 | Threshold-flag performance (recall 1.00, precision 0.73) accepted as-is, not further tuned | Tune threshold for higher precision | Zero false negatives means no genuinely at-risk student is missed; the 10 false positives only cost advisor time on students who didn't need it. Given the intervention-support business context, missing a struggling student is a worse outcome than over-flagging — current threshold behavior fits the stated business need | Confusion matrix above |

Final Model Comparison & Recommendation

Recommended model: Gradient Boosting (test RMSE 1.945, R^2 0.816) — selected over Random Forest despite RF's better CV score, due to RF's clear overfitting (train-test gap 1.415 vs GB's 0.985). See decision #13.

Top predictors (2-method agreement): G2, absences, G1 — in that order, confirmed by both tree-based and permutation importance (decision #17).

Key limitations documented:

- Model's strength depends heavily on G1/G2 — not a true start-of-year predictor, only useful from second-period onward.
- Random Forest's CV-vs-test disagreement shows CV score alone is not a reliable model-selection criterion on a dataset this small (316 train rows).
- Trained on Math dataset only (395 rows) — generalizability to the Portuguese-language dataset or other institutions is unverified.
- Per-student explanations are simplified (global top features per student, not individualized SHAP-style attribution).
- Small dataset limits statistical confidence of all reported metrics.